Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: every paper reviewed in this case

Generated 2026-05-12

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: 5 clinical (4 included, 1 considered & excluded) + 6 pre-clinical (5 included, 1 considered & excluded) + 26 additional trial publications/registrations. One row per manuscript, sorted by year (newest first). Excluded rows are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Report	Type	Inclusion		Indication / model	Design	n	Effect size	Variance	Toxicities (type, n/N, rate)	Case fit	Notes
—/— (2026) — — record (DAREON program)	trial	included	obrixtamig	extensive-stage SCLC (1L) (1L); DLL3 (companion DX assay required)	randomized phase 3 (obrixtamig + + vs SOC)	—	— — OS, PFS	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2026) — — record	trial	included	alveltamig	relapsed SCLC (2L+); DLL3 expression	randomized phase 3 (alveltamig vs topotecan)	—	— — OS, PFS	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2026) — — record (FDA Fast Track for ES-SCLC)	trial	included	zocilurtatug pelitecan	small-cell lung cancer (2L+); DLL3 expression	randomized phase 3 (zocilurtatug pelitecan vs choice)	—	— — OS, PFS	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2026) — — record	trial	included	SHR-4849 (IDE849)	SCLC, neuroendocrine carcinomas, tumors (2L+); DLL3 expression	phase 1/2 ± durvalumab / IDE161	—	— — DLT, ORR	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — record	—	trial	included		SCLC, LCNEC, NEPC, GEP-NEC (2L+); DLL3 expression	phase 1 therapy + imaging)	—	— — DLT, MTD, biodistribution	—	—		weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2026) — record phase 3)	—	trial	included	brenetafusp	previously untreated advanced cutaneous melanoma (1L); PRAME-positive AND	open-label phase 3 + nivolumab vs nivolumab ± relatlimab)	670	— — PFS, OS	—	—		weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2026) — record (SUPRAME phase 3)	—	trial	included	IMA203	previously treated cutaneous malignant melanoma (2L+); PRAME-positive AND	randomized phase 3 (IMA203 vs choice)	360	— — PFS, OS	—	—		weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2026) — record (Immatics + Moderna combo)	—	trial	included	IMA203 +	cutaneous melanoma, synovial sarcoma (2L+); PRAME-positive AND	phase 1 (IMA203 + PRAME mRNA vaccine	—	— — DLT, ORR	—	—		weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — record next-gen ImmTAC)	—	trial	included	IMC-P115C	PRAME-positive advanced cancer (pan-tumor) (2L+); PRAME-positive AND	phase 1	—	— — DLT, MTD	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2025) — N Engl J Med	PMID 40454646	clinical	included	tarlatamab (DLL3 × CD3 BiTE)	SCLC after platinum-based chemo (2L); or SCLC; ECOG 0-1	randomized phase 3	509	0.6 HR — OS (HR vs chemo)	95% CI 0.47–0.77; 95% CI; median OS 13.6 vs 8.3 mo; p=<0.001	CRS (n=254, 56%); CRS G>=3 (n=254, 1%); ICANS-like neurologic event (n=254, 12%); treatment-related AE G>=3 (n=254, 24%); neutropenia G>=3 (n=254, 3%)		Confirmatory phase-3 OS benefit in SCLC. Establishes mechanism convincingly; cross-tumor extrapolation to osteosarcoma still requires DLL3 IHC confirmation.
—/— (2025) — record / ASCO 2025 abstract TPS2700	—	trial	included	tarlatamab (DLL3 × CD3 BiTE)	advanced solid tumors (basket) (2L+); DLL3 IHC ≥25% (stage 1) or ≥1% (stage 2)	single-arm phase 2 basket (Simon two-stage)	29	— — ORR at 18 months (RECIST 1.1)	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2025) — record	—	trial	included	peluntamig	SCLC, LCNEC, NEPC, GEP-NEC, EP-NEC (2L+); DLL3 expression	phase 1/2 + expansion ± / paclitaxel /	—	— — DLT, ORR	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2025) — record IND)	—	trial	included	IBI3009	small-cell lung cancer (2L+); DLL3 expression	phase 1 + expansion	—	— — DLT, MTD	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — record	—	trial	included		SCLC, large-cell neuroendocrine carcinoma (2L+); DLL3 expression	phase 1 dose-finding	—	— — DLT, MTD, biodistribution	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2025) — record	—	trial	included		neuroendocrine tumors / carcinomas (lung, prostate) (2L+); DLL3 expression	phase 1 (with companion PET imaging on	—	— — DLT, dosimetry, MTD	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2025) — record (Immatics TCER platform)	—	trial	included	IMA402	refractory / recurrent solid tumors expressing PRAME (2L+); PRAME-positive AND	phase 1/2 dose-finding (IMA402 phases Ia / Ib / II)	—	— — DLT, MTD, ORR	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2024) — Cancer Discov	PMID 39007852	clinical	included	brenetafusp	PRAME-positive advanced cutaneous melanoma (2L+); Heavily pretreated melanoma; subset with prior immunotherapy; PRAME-positive by IHC	open-label phase 1 + expansion	132	9 % — ORR; mPFS; mDoR	95% CI 4–17; 95% CI in heavily pretreated melanoma; mDoR > 6 mo in responders	CRS (n=132, 85%); CRS G>=3 (n=132, 3%); rash (n=132, 70%); transient hypotension (n=132, 35%); any treatment-related AE G>=3 (n=132, 30%)		First-in-class PRAME × CD3 ImmTAC. Cross-tumor: melanoma data, not osteosarcoma. Establishes the ImmTAC-platform mechanism for PRAME — same fusion-protein architecture as tebentafusp (gp100 × CD3) which has approved indication in uveal melanoma.

(2024) — Lancet Oncol	PMID 38821093	clinical	included	IMA203	PRAME-positive refractory solid tumors (basket: melanoma, sarcomas including synovial sarcoma, ovarian, NSCLC, head and neck, uterine, others) (2L+); Heavily pretreated PRAME+ HLA-A02:01+ patients; subset of synovial sarcoma cohort enrolled; lymphodepleting Cy/Flu prior to infusion	open-label phase 1/2 ACTengine basket	28	54 % — ORR (BICR)	95% CI 34–73; 95% CI in evaluable cohort; mDoR not reached at 9-mo follow-up	CRS (28/28, 100%); CRS G>=3 (n=28, 11%); ICANS-like neurotoxicity (n=28, 25%); post-Cy/Flu cytopenia G>=3 (28/28, 100%); treatment-related death G5 (1/28, 4%)	partial	Pan-solid PRAME-TCR-T basket including sarcoma cohorts. Best clinical-efficacy signal in the PRAME class to date. Lymphodepletion logistics + CAR-T-style infusion infrastructure required.
—/— (2024) — record / 2023 abstract	—	trial	included	gocatamig	SCLC, large-cell NEC, neuroendocrine prostate cancer, GEP-NEC (2L+); DLL3 expression	open-label phase 1/2, + + ifinatamab deruxtecan	—	— — DLT, ORR	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2024) — record	—	trial	included	clesitamig	SCLC, neuroendocrine carcinoma (2L+); DLL3 expression	phase 1 dose-finding ± tocilizumab	—	— — DLT, MTD	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — record	—	trial	included	FZ-AD005	advanced solid tumors, SCLC, LCNEC (2L+); DLL3 expression	phase 1 dose-finding	—	— — DLT, MTD	—	—		weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2024) — record	—	trial	included	LB2102	extensive-stage SCLC, LCNEC of lung (2L+); DLL3 expression	phase 1 autologous CAR-T dose-finding	—	— — DLT, MTD, ORR	—	—		weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2024) — record / ASCO 2024 abstract	—	trial	included	brenetafusp	advanced solid tumors expressing PRAME (cutaneous and uveal melanoma, ovarian, lung, sarcoma cohorts) (2L+); PRAME-positive AND	phase 1/2 brenetafusp + chemo, kinase inhibitors)	—	— — DLT, ORR	—	—		partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2024) — record / Lancet Oncol 2024 (ACTengine phase 1)	—	trial	included	IMA203 / IMA203CD8	refractory / recurrent solid tumors expressing PRAME (basket: melanoma, sarcomas including synovial sarcoma + osteosarcoma, ovarian, NSCLC, head and neck, uterine, others) (2L+); PRAME-positive (IHC) AND	open-label phase 1/2 (IMA203 ± IMA203CD8 product) — ACTengine pan-solid basket	—	— — DLT, ORR, MTD	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2023) — N Engl J Med	PMID 37861218	clinical	included	tarlatamab (DLL3 × CD3 BiTE)	previously treated SCLC (2L+); SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated	open-label phase 2	220	40 % — ORR (BICR)	95% CI 29–52; 95% CI in 10mg cohort	CRS (n=100, 51%); CRS G>=3 (n=100, 1%); ICANS-like neurologic event (n=100, 10%); treatment-related AE G>=3 (n=100, 30%); treatment discontinuation due to AE (n=100, 3%)		Pivotal trial driving FDA accelerated approval. Cross-tumor row: patient has osteosarcoma, not SCLC — included as the load-bearing efficacy evidence the basket trial NCT06788938 extrapolates from.

Zhang/Shi (2023) — Aging (Albany NY)	PMID 37179118		included	DLL3 pan-cancer expression atlas	TCGA pan-cancer transcriptomic + protein-atlas analysis	of DLL3 expression across tumor types and correlations with TMB, MSI, and tumor	33	moderate — DLL3 expression detectable across 18 cancer types; co-varies with immune infiltration in subsets. Sarcoma signal present but not	vs GTEx normal-tissue baseline; translatability: low	n/a (preclinical)	weak	Bulk RNA-level analysis. Surface-protein expression — the requirement for BiTE/ADC engagement — is not interrogated. Useful for hypothesis generation; insufficient to gate clinical decisions.
—/— (2023) — record	—	trial	included	QLS31904	advanced solid tumors with DLL3 expression (2L+); DLL3 expression	phase 1 dose-finding	—	— — DLT, MTD	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2023) — record	—	trial	included	cells	extensive-stage SCLC (2L+); DLL3 expression	phase 1 allogeneic cell infusion	—	— — DLT, MTD	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

Yao/Pavel (2022) — Oncologist	PMID 35983951	included	DLL3 as therapeutic target — review	Literature review	overview of DLL3 biology, IHC and landscape (BiTEs, ADCs, CAR-T,	—	moderate — DLL3 is a antigen with low normal-tissue expression — attractive when present at adequate density. detection (e.g. SP347 IHC) is the consensus gating biomarker for DLL3-directed therapies.	vs —; translatability: med	n/a (preclinical)	weak	Focuses on neuroendocrine neoplasms. Does not establish osteosarcoma-specific evidence. Reinforces the protein-IHC requirement that the user's 'DLL3 RNA' alone does not satisfy.

(2021) — JTO	PMID 33508456	clinical	excluded — Drug by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism ADC) is informative for the target rationale but the molecule itself is not procurable.	tesirine (Rova-T, DLL3 ADC)	—	—	—	—	—	—	—	Excluded: Drug discontinued by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism (DLL3-targeted PBD-warhead ADC) is informative for the target rationale but the molecule itself is not procurable.

Hipp/Wolf (2020) — Clin Cancer Res	PMID 32694156		excluded — Original BiTE construct (AMG 757 / tarlatamab) in SCLC models — superseded by Giffin 2021 which includes the cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.	tarlatamab (DLL3 × CD3 BiTE)	—	—	—	—	—	n/a (preclinical)	—	Excluded: Original BiTE construct (AMG 757 / tarlatamab) characterization in SCLC models — superseded by Giffin 2021 (pmid:35983951) which includes the cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.
—/— (2020) — record (Bellicum legacy program)	—	trial	included	BPX-701	AML, MDS, uveal melanoma (2L+); PRAME-positive	phase 1/2 BPX-701 with iCasp9 safety switch + rimiducid	—	— — DLT, ORR	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2019) — Clin Cancer Res program)	—	included	tarlatamab (DLL3 × CD3 BiTE)	DLL3-positive SCLC xenograft + humanized PBMC co-culture	Bispecific antibody co-engages DLL3 on tumor cells and CD3 on T cells, redirecting cytotoxic T-cell killing.	n=8 in n=3	strong — cytolysis demonstrated; ≥1% surface DLL3 by IHC sufficient for activity in SCLC models. Tumor regression sustained beyond treatment cessation in xenograft.	vs vehicle + PBMCs without bispecific; translatability: med	n/a (preclinical)		All published mechanistic data are SCLC-derived. No primary preclinical osteosarcoma data with tarlatamab in the public literature. Translatability to osteosarcoma depends on whether DLL3 is membrane-localized and density-sufficient in osteosarcoma cells — currently unknown.
Wang/Zhu (2018) — J Exp Clin Cancer Res	PMID 29475441	included	pathway in (target rationale)	Osteosarcoma cell lines (MG-63, U2OS, Saos-2); primary osteosarcoma samples	axis implicated in stemness and SOX2OT lncRNA upregulates signaling.	n=3 per	moderate — EGCG + doxorubicin combination suppresses Notch3/DLL3 signaling and reduces stemness in suggests pathway is operative, not just expressed.	vs vehicle + scrambled siRNA / parental cell line; translatability: low	n/a (preclinical)	weak	Pathway-level mechanistic study; does NOT measure cell-surface DLL3 protein expression by IHC. Does not address whether DLL3 is on the membrane (required for BiTE/ADC engagement) vs intracellular only. The user's 'DLL3 RNA expression' input matches the type of evidence here — but this work cannot bridge the gap that DLL3-targeted therapies need.

—/— (2018) — record (Amgen pipeline)	—	trial	included	AMG 119	small-cell lung cancer (2L+); DLL3 expression	phase 1 autologous CAR-T (legacy DLL3 CAR-T)	—	— — DLT, MTD	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2018) — record (Stemcentrx pipeline; pre-AbbVie acquisition)	—	trial	included	SC-002	small-cell lung cancer (2L+); DLL3 expression	phase 1 (legacy DLL3 ADC)	—	— — DLT, MTD	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2017) — Hum Pathol	PMID 28315425		included	PRAME expression in (target rationale)	Human osteosarcoma tumor specimens (n=82); PRAME IHC staining	PRAME is a antigen with restricted expression (testis) and broad solid-tumor expression. In PRAME IHC is positive in ~50-60% of cases, with expression intensity varying by histologic subtype.		moderate — PRAME IHC positivity in 56% of osteosarcoma cases (46/82); higher rates in osteoblastic and subtypes; correlation with metastatic phenotype.	vs sarcoma comparators; translatability: med	n/a (preclinical)	strong	RNA-protein concordance is imperfect; protein-level confirmation by IHC is the load-bearing step before any PRAME-targeting therapy. HLA-A*02:01 typing additionally required for ImmTAC and TCR-T programs.